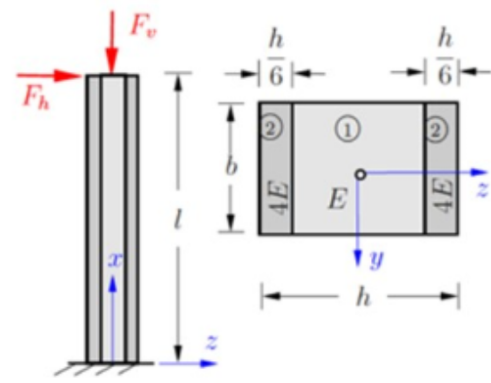


Homework 4.11 A column is clamped at the bottom and is carrying a vertical load F_v at the center of the top cross section and a horizontal load F_h in the middle of edge b . The column is made of 3 layers with different Young's moduli. Determine (1) the normal stress distribution in the cross section at the clamping; (2) the shear stress in the interface between the two materials at the clamping.



(1) From equilibrium conditions:

$$-F_v = \sigma_1 A_1 + \sigma_2 A_2$$

$$A_1 = \frac{2}{3}hb \quad A_2 = 2\left(\frac{h}{6}\right)b = \frac{1}{3}hb$$

From Hooke's law, $\sigma = E\varepsilon$

$$\sigma_1 = E_1 \varepsilon_1 \quad \sigma_2 = E_2 \varepsilon_2$$

$$\text{since } \varepsilon_1 = \varepsilon_2 = \varepsilon$$

$$-F_v = E_1 \varepsilon A_1 + E_2 \varepsilon A_2$$

$$\sigma_1 = E_1 \varepsilon = E \cdot -\frac{F_v}{2Ehb} = -\frac{F_v}{2hb}$$

$$-F_v = E\varepsilon\left(\frac{2}{3}hb\right) + 4E\varepsilon\left(\frac{1}{3}hb\right)$$

$$\sigma_2 = E_2 \varepsilon = 4E \cdot -\frac{F_v}{2Ehb} = -2\frac{F_v}{hb}$$

$$-F_v = \frac{2}{3}E\varepsilon hb + \frac{4}{3}E\varepsilon hb$$

$$-F_v = 2E\varepsilon hb$$

$$\varepsilon = -\frac{F_v}{2Ehb}$$

$$M_s = -F_h l \quad \sigma = E(z) \varepsilon(z) \quad \varepsilon(z) = \psi' z$$

$$\begin{aligned} M = \int \sigma z dA &= 2b\psi' \left[E_1 \int_{-h/6}^{h/6} z^2 dz + E_2 \int_{h/6}^{h/3} z^2 dz \right] \\ &= 2b\psi' \left\{ E \left[\frac{z^3}{3} \right]_{-h/6}^{h/6} + 4E \left[\frac{z^3}{3} \right]_{h/6}^{h/3} \right\} \\ &= 2b\psi' E \left[\frac{h^3}{3^4} + \frac{4}{3} \left[\frac{h^3}{8} - \frac{h^3}{27} \right] \right] \\ &= \frac{7}{27} b\psi' E h^3 \end{aligned}$$

$$M = M_s \quad -F_h l = \frac{7}{27} b\psi' E h^3$$

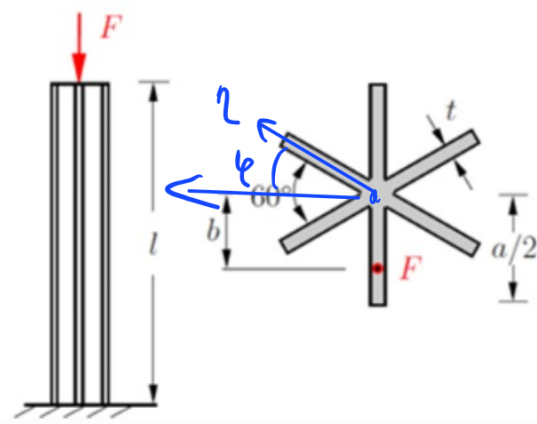
$$\psi' = -\frac{27}{7} \frac{F_h l}{b^4 E h^3}$$

$$\sigma_1 = E_1 \psi' z = E \cdot -\frac{27}{7} \frac{F_h l}{b^4 E h^3} \cdot z = -\frac{27}{7} \frac{F_h l z}{b^4 h^3}$$

$$\sigma_2 = E_2 \psi' z = 4E \cdot -\frac{27}{7} \frac{F_h l}{b^4 E h^3} \cdot z = -\frac{108}{7} \frac{F_h l z}{b^4 h^3}$$

Homework 4.12 The column with a star-shaped cross section ($t \ll a$) is loaded by a force F , applied off center. Determine

- the moments of inertia
- the maximum absolute value of the stress,
- the maximal value of b such that nowhere in the cross section tensile stresses occur.



a)
$$I_y = \frac{ta^3}{12}$$

$$I_\eta = \frac{at^3}{12} \quad I_\zeta = \frac{ta^3}{12} \quad \varphi = -30^\circ \quad I_{\eta\zeta} = 0$$

$$\begin{aligned} I_{y\eta} = I_{\eta y} &= \frac{I_\eta I_\zeta}{2} + \frac{I_\eta - I_\zeta}{2} \cos 2\varphi + I_{\eta\zeta} \sin 2\varphi \\ &= \frac{ta^3}{24} + \frac{at^3 - ta^3}{24} \left(\frac{1}{2}\right) + 0 \\ &= \frac{ta^3}{24} - \frac{ta^3}{48} = \frac{ta^3}{48} \end{aligned}$$

$$I_y = I_{yI} + 2I_{y\eta} = \frac{ta^3}{12} + \frac{ta^3}{24} = \frac{ta^3}{8}$$

b)
$$\sigma = \frac{N}{A} + \frac{My}{I_y} z = -\frac{F}{3at} - \frac{8bt}{ta^3} z$$

Maximum stress at $z = a/2$

$$\sigma_{\max} = -\frac{F}{3at} - \frac{4bF}{ta^2}$$

$$\begin{aligned} &= -\frac{F}{3at} - \frac{4bF}{ta^2} \\ &= -\frac{F}{at} \left(\frac{1}{3} + \frac{4b}{a} \right) \end{aligned}$$

c) Tensile stress occurs first at $z = -a/2$

$$\sigma\left(-\frac{a}{2}\right) \geq 0 \quad -\frac{F}{3at} - \frac{8bt}{ta^3} \left(-\frac{a}{2}\right) \geq 0$$

$$\frac{F}{3at} = \frac{4bF}{ta^2} \quad b \geq \frac{a}{12}$$